# Md Shadnan Azwad Khan

Curriculum Vitæ

*Last update: November 3, 2024.*

## Personal Information


**Nationality:** Bangladeshi **Date of Birth:** February 7, 1994

**Email:** shadnan.azwad@proton.me **Phone:** +33 7 51 31 47 60 **LinkedIn:** Shadnan Azwad Khan


**Objective:** To pursue a research internship (master's thesis) in quantum information theory, quantum computing, and quantum machine learning.

## Education

2023 – PRESENT

**Master of Science in Computer Science — Quantum Information**,
*Sorbonne Université*, Paris, France.

**Master Year 2**: ongoing.

**Master Year 1**: Mention Bien.

**4EU+ Alliance Shared Course**: "Quantum Information and Quantum Many-Body Theory", with summer school at the *University of Copenhagen*, Copenhagen, Denmark. Completed as a part of the Advanced Quantum Information Project during Master Year 1.

2013—2018

**Bachelor of Science in Electrical and Electronic Engineering with a second major in Computer Science**,
*BRAC University*, Dhaka, Bangladesh.

Graduated with a double major and honours (academic distinction).

**Final Year Design Project Title**: "Simultaneous localization and mapping (SLAM) with an autonomous robot."

**Final Year Research Project Title:** "A new multi-robot search algorithm using probabilistic finite state machine and Lennard-Jones potential function."

**Research Semester Abroad**: Erasmus+ visiting student, *University of Staffordshire*, Stoke-on-Trent, United Kingdom. Completed as a part of the Final Year Research Project.

2012

**High School Diploma**,
*The Aga Khan School, Dhaka*, Dhaka, Bangladesh.

Completed GCE Advanced levels (attaining Grade *A* in Physics, Mathematics, and Chemistry) under the University of Cambridge International Education curriculum.

## Experience

MAR 2019 – AUG 2023

**Research Assistant**,
*Independent University, Bangladesh (IUB)*, Dhaka, Bangladesh.

**Research Areas**: Swarm Intelligence, Unsupervised Machine Learning, Network Anomaly Detection, and Health Informatics.

JUL 2012 – NOV 2012

**Substitute Information and Communications Technology Teacher**,
*The Aga Khan School, Dhaka*, Dhaka, Bangladesh.

**Duties:** Academic lesson planning, classroom instruction, and laboratory supervision.

## Honours, Grants & Awards

2024—2025 **QICS scholarship grant**, *Quantum Information Center Sorbonne, Sorbonne Université*

2023—2024 **Maîtrise (Master 1) completed with Mention Bien**, *Sorbonne Université*

2024 **4EU+ short-term mobility grant**, *4EU+ Alliance*

2024 **Solidarity scholarship**, *Groupe des Anciens, Sorbonne Université*

2018 **Bachelor's completed with Honours (academic distinction)**, *BRAC University*

2017 **Learning mobility grant for study abroad programme at University of Staffordshire**, *Erasmus+ Key Action 1 programme*

2015 **Dean's list (Summer 2015)**, *BRAC University*

2013, 2014 **Vice-Chancellor's list (Fall 2013, Summer 2014)**, *BRAC University*

2010, 2012 **The Daily Star Award for outstanding achievement in Advanced Level and Ordinary Level Examinations**, *The Daily Star*

## Publications

**Google Scholar**: Shadnan Azwad Khan **ORCiD**: 0000-0003-2769-6856

*Previously published under the name Muhammad S. A. Khan (Md. is an abbreviation of Muhammad).*

### Peer-Reviewed Publications

[1] Qi Huang, Emanuele Mezzi, Osman Mutlu, Miltiadis Kofinas, Vidya Prasad, **Shadnan Azwad Khan**, Elena Ranguelova, and Niki van Stein. Beyond the veil of similarity: Quantifying semantic continuity in explainable AI. In *Explainable Artificial Intelligence. xAI 2024. Communications in Computer and Information Science*, volume 2153, 2024.

DOI: 10.1007/978-3-031-63787-2_16

[2] **Muhammad S. A. Khan**, Tarem Ahmed, and Mohammad Faisal Uddin. Multi-robot search algorithm using timed random switching of exploration approaches. In *2020 IEEE Region 10 Symposium (TENSYMP)*, pages 868–871, 2020.

DOI: 10.1109/TENSYMP50017.2020.9230829

[3] **Muhammad S. A. Khan**, Mohammad S. Hasan, and Tarem Ahmed. A new multi-robot search algorithm using probabilistic finite state machine and Lennard-Jones potential function. In *2018 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, pages 850–855, 2018.

DOI: 10.1109/ROBIO.2018.8665082

[4] **Muhammad S. A. Khan**, Shoumik S. Chowdhury, Nafis Niloy, Fatema Tuz Zohra Aurin, and Tarem Ahmed. Sonar-based slam using occupancy grid mapping and dead reckoning. In *TENCON 2018 - 2018 IEEE Region 10 Conference*, pages 1707–1712, 2018.

DOI: 10.1109/TENCON.2018.8650124

### In Preparation

[5] **Shadnan Azwad Khan**, Tarem Ahmed, Salva Daneshgadeh Çakmakçı. Systematic review of unsupervised learning techniques in anomaly-based network intrusion detection systems. Abstract available online.

[6] Nazifa M. Shemonti, **Shadnan Azwad Khan**, Shifat Uddin, Saifur Rahman, Tarem Ahmed, Mohammad Faisal Uddin. Monitoring patient deterioration using unsupervised kernel-based online anomaly detection. Abstract available online.

## Presentations & Workshops

[1] **Hidden Nonlocality**

Poster presentation (available online), 4EU+ Quantum Information and Quantum Many-Body Theory Summer School in Copenhagen.

[2] **Investigating Quantum Reservoir Computing for Time Series Forecasting**

Poster presentation (available online), Paris Centre for Quantum Technologies (PCQT) Workshop. Completed during Master Year 1 for the Quantum Information Group Project. Project report available online.

[3] **Do machines learn non-explicit semantics?**

Case study, ICT with Industry 2024, Lorentz Center. Results published at the 2nd World Conference on eXplainable Artificial Intelligence. Code available on Github.

## Projects

**Github:** shadwad ~ *Includes additional projects and recent updates.*

[1] **Quantum Digital Payment**

Challenge, Pan-European Quantum Internet Hackathon 2024. Code available on GitHub.

[2] **Complex Arithmetic and FFT-based Polynomial Multiplication in C**

Course Project, Numerical and Symbolic Algorithms Modeling, Sorbonne Université. Available on GitHub. Project report available online.

## Skills

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|---|---|
| NATURAL LANGUAGES | English (Fluent, C2 Proficiency) · Bangla (Fluent, Native Proficiency) French (Beginner, A1 Proficiency) |
| PROGRAMMING | Python (NumPy · TensorFlow · PyTorch) · Java · LaTeX · C/C++ · MATLAB Mathematica · SageMath · Maple |